Fluid Mechanics

RV College of Engineering

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Symbols used

Symbols	
V_m	Volume of the fluid
V_ρ	Specific volume of the fluid
u	x component of velocity of the fluid at a point
v	y component of velocity of the fluid at a point
w	z component of velocity of the fluid at a point
V	Magnitude of velocity of the fluid at a point

Classical Physics for Engineers
Mechanical Engineering Stream – (ME, AS, CH, IM)
Unit-IV Fluid Mechanics

1 Definition of fluid

A fluid is a substance that can flow. Fluids conform to the boundaries of any container in which we put them. They do so because a fluid cannot sustain a force that is tangential to its surface. In the more formal language, a fluid is a substance that flows because it cannot withstand a shearing stress. It can, however, exert a force in the direction perpendicular to its surface. Both liquids and gases are categorized as fluids.

Fluids generally fall into two classes, liquids and gases. A gas always expands to fill the entire volume of its container. In contrast, the volume of a liquid changes little, so that it may not completely fill a large container.

A substance exists in three primary phases: solid, liquid, and gas. (At very high temperatures, it also exists as plasma.) A substance in the liquid or gas phase is referred to as a fluid. Distinction between a solid and a fluid is made on the basis of the substance's ability to resist an applied shear (or tangential) stress that tends to change its shape. A solid can resist an applied shear stress by deforming, whereas *a fluid deforms continuously under the influence of a shear stress*, no matter how small. In solids, stress is proportional to *strain*, but in fluids, stress is proportional to *strain rate*. When a constant shear force is applied, a solid eventually stops deforming at some fixed strain angle, whereas a fluid never stops deforming and approaches a constant *rate* of strain.

2 Concept of continuum

A fluid is composed of molecules which may be widely spaced apart, especially in the gas phase. Yet it is convenient to disregard the atomic nature of the fluid and view it as continuous, homogeneous matter with no holes, that is, a continuum. The continuum idealization allows us to treat properties as point functions and to assume that the properties vary continually in space with no jump discontinuities. This idealization is valid as long as the size of the system we deal with is large relative to the space between the molecules. This is the case in practically all problems, except some specialized ones. The continuum idealization is implicit in many statements we make, such as "the density of

water in a glass is the same at any point."

To have a sense of the distances involved at the molecular level, consider a container filled with oxygen at atmospheric conditions. The diameter of an oxygen molecule is about 3×10^{-10} m and its mass is 5.3×10^{-26} kg. Also, the *mean free path* of oxygen at 1 atm pressure and 20°C is 6.3×10^{-8} m. That is, an oxygen molecule travels, on average, a distance of 6.3×10^{-8} m (about 200 times its diameter) before it collides with another molecule.

Also, there are about 3×10^{16} molecules of oxygen in the tiny volume of $1\,mm^3$ at 1 atm pressure and 20°C. The continuum model is applicable as long as the characteristic length of the system (such as system's length, breadth or diameter) is much larger than the mean free path of the molecules. At very low pressure, e.g., at very high elevations, the mean free path may become large (for example, it is about 0.1 m for atmospheric air at an elevation of 100 km). For such cases the impact of individual molecules should be considered. In this text we limit our consideration to substances that can be modelled as a continuum.

3 Classification of fluids

- **Ideal fluid:** A fluid, which is incompressible and having no viscosity, is known as an ideal fluid. Ideal fluid is only an imaginary fluid as all the fluids, which exist, have some viscosity.
- **Real fluid:** A fluid, which possesses viscosity, is known as real fluid. All the fluids, in actual practice, are real fluids. Example: Water, Air etc.
- **Newtonian fluid:** A real fluid, in which shear stress is directly proportional to the rate of shear strain or velocity gradient, is known as a Newtonian fluid. Example: Water, Benzine etc.
- **Non Newtonian fluid:** A real fluid, in which shear stress in not directly proportional to the rate of shear strain or velocity gradient, is known as a Non Newtonian fluid. Example: Plaster, Slurries, Pastes etc.
- **Ideal plastic fluid:** A fluid, in which shear stress is more than the yield

value and shear stress is proportional to the rate of shear strain or velocity gradient, is known as ideal plastic fluid.

- **Incompressible fluid:** A fluid, in which the density of fluid does not change with the change in external force or pressure, is known as incompressible fluid. All liquids are considered in this category.
- **Compressible fluid:** A fluid, in which the density of fluid changes while change in external force or pressure is known as compressible fluid. All gases are considered in this category.

4 Fluid Properties

Any characteristic of a system is called a property. Some familiar properties are pressure P, temperature T, volume V_m, and mass m. The list can be extended to include less familiar ones such as viscosity, thermal conductivity, modulus of elasticity, thermal expansion coefficient, electric resistivity, and even velocity and elevation.

1. Mass and Weight Properties
   (a) Density (ρ): Ratio of mass to volume at a point.
   (b) Specific Weight (γ): Ratio of weight to volume at a point
   (c) Specific Gravity (S or SG): Ratio of (density of a liquid) to (density of water at 4°C)
2. Properties Related to Viscosity
   (a) Viscosity (μ): A property that characterizes resistance to shear stress and fluid friction
   (b) Kinematic Viscosity (ν): A property that characterizes the mass and viscous properties of a fluid
3. Miscellaneous Properties

(a) Surface Tension (σ): A property that characterizes the tendency of a liquid surface to behave as a stretched membrane

(b) Vapour Pressure P_v: The pressure at which a liquid will boil

(c) Bulk Modulus of Elasticity κ: A property that characterizes the compressibility of a fluid

4.1 Density

Density is defined as mass per unit volume.

$$\rho = \frac{m}{V_m} \qquad (\text{in kg}/m^3)$$

where m is the mass of the fluid and V_m is the volume of the fluid.

The reciprocal of density is the *specific volume* V_ρ which is defined as *volume per unit mass.* That is,

$$V_\rho = \frac{V_m}{m} = \frac{1}{\rho}.$$

For a differential volume element of mass δm and volume δV_m, density can be expressed as

$$\rho = \frac{\delta m}{\delta V_m}.$$

The density of a substance, in general, depends on temperature and pressure. The density of most gases is proportional to pressure and inversely proportional to temperature. Liquids and solids, on the other hand, are essentially incompressible substances, and the variation of their density with pressure is usually negligible. At 20°C, for example, the density of water changes from 998 kg/m^3 at 1 atm to 1003 kg/m^3 at 100 atm, a change of just 0.5 percent. The density of liquids and solids depends more strongly on temperature than it does on pressure. At 1 atm, for example, the density of water changes from 998 kg/m^3 at 20°C to 975 kg/m^3 at 75°C, a change of 2.3 percent, which can still be neglected in many engineering analyses.

4.1.1 Density of Ideal Gases

Property tables provide very accurate and precise information about the properties, but sometimes it is convenient to have some simple relations among the properties that are sufficiently general and reasonably accurate. Any equation that relates the pressure, temperature, and density (or specific volume) of a substance is called an *equation of state*. The simplest and best-known equation of state for substances in the gas phase is the *ideal-gas equation of state*, expressed as

$$PV_m = nR_uT$$

where P is the pressure, V_m is the volume, n is the number of moles, $R_u = 8.314\,\text{kJ}\cdot\text{kmol}^{-1}\cdot\text{K}^{-1}$ is the universal gas constant, T is the temperature of the ideal gas.

If M is the molar mass (also called molecular weight) of the gas molecules then $m = nM$ is the total mass of the gas. Therfore the above equation can be written as

$$PV_m = \frac{nM}{M}R_uT = m\frac{R_u}{M}T = mRT$$

where R is the gas constant. The gas constant R is different for each gas and is determined from $R = R_u/M$.

Rearranging the above equation with $\rho = m/V_m$ as its density we get,

$$P = \frac{m}{V_m}RT = \rho RT$$

4.2 Specific gravity, or relative density

Sometimes the density of a substance is given relative to the density of a well-known substance. Then it is called specific gravity, or relative density, and is defined as *the ratio of the density of a substance to the density of some standard substance at a specified temperature* (usually water at 4°C, for which $\rho_w = 1000\ \text{kg/m}^3$). That is,

$$\text{SG} = \frac{\rho}{\rho_w}$$

Note that the specific gravity of a substance is a dimensionless quantity.

4.3 Specific weight

The weight of a unit volume of a substance is called specific weight or weight density and is expressed as

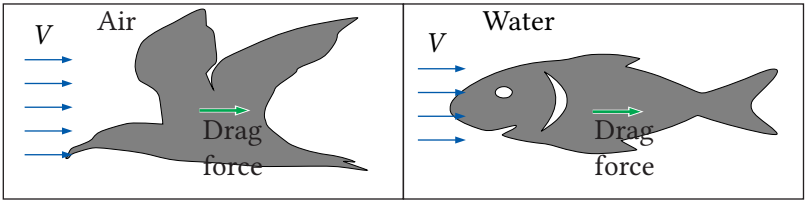
$$\gamma_s = \rho g \qquad (\mathrm{N/m^3})$$

where g is the gravitational acceleration.

4.4 Newton's Law of viscosity

When two solid bodies in contact move relative to each other, a friction force develops at the contact surface in the direction opposite to motion. To move a table on the floor, for example, we have to apply a force to the table in the horizontal direction large enough to overcome the friction force. The magnitude of the force needed to move the table depends on the *friction coefficient* between the table legs and the floor.

The situation is similar when a fluid moves relative to a solid or when two fluids move relative to each other. We move with relative ease in air, but not so in water. Moving in oil would be even more difficult, as can be observed by the slower downward motion of a glass ball dropped in a tube filled with oil. It appears that there is a property that represents the internal resistance of a fluid to motion or the "fluidity," and that property is the *viscosity*. The force a flowing fluid exerts on a body in the flow direction is called the *drag force*, and the magnitude of this force depends, in part, on viscosity.

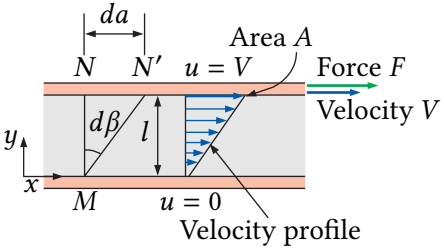


To obtain a relation for viscosity, consider a fluid layer between two very large parallel plates (or equivalently, two parallel plates immersed in a large body of a fluid) separated by a distance l. Now a constant parallel force F is

applied to the upper plate while the lower plate is held fixed. After the initial transients, it is observed that the upper plate moves continuously under the influence of this force at a constant speed V. The fluid in contact with the upper plate sticks to the plate surface and moves with it at the same speed, and the shear stress τ acting on this fluid layer is

$$\tau = \frac{F}{A}$$

where A is the contact area between the plate and the fluid. Note that the fluid layer deforms continuously under the influence of shear stress.



The fluid in contact with the lower plate assumes the velocity of that plate, which is zero (because of the no-slip condition). In steady laminar flow, the fluid velocity between the plates varies linearly between 0 and V, and thus the *velocity profile* (u) and the *velocity gradient* are

$$u(y) = \frac{y}{l}V \qquad \text{and} \qquad \frac{du}{dy} = \frac{V}{l}$$

where y is the vertical distance from the lower plate.

During a differential time interval dt, the sides of fluid particles along a vertical line MN rotate through a differential angle $d\beta$ while the upper plate moves a differential distance $da = Vdt$. The angular displacement or deformation (or shear strain) can be expressed as

$$d\beta \approx \tan d\beta = \frac{da}{l} = \frac{Vdt}{l} = \frac{du}{dy}dt$$

Rearranging, the rate of deformation under the influence of shear stress τ becomes

$$\frac{d\beta}{dt} = \frac{du}{dy}$$

Thus we conclude that the rate of deformation of a fluid element is equivalent to the velocity gradient du/dy. Further, it can be verified experimentally that for most fluids the rate of deformation (and thus the velocity gradient) is directly proportional to the shear stress τ,

$$\tau \propto \frac{d\beta}{dt} \qquad \text{and hence} \qquad \tau \propto \frac{du}{dy}$$

Fluids for which the rate of deformation is linearly proportional to the shear stress are called *Newtonian fluids* after Sir Isaac Newton, who expressed it first in 1687. Most common fluids such as water, air, gasoline, and oils are Newtonian fluids. Blood and liquid plastics are examples of non-Newtonian fluids.

In one-dimensional shear flow of Newtonian fluids, shear stress can be expressed by the linear relationship

$$\tau = \mu \frac{du}{dy} \qquad (\mathrm{N/m^2}).$$

This expression is referred to as *Newton's law of viscosity*. The constant of proportionality μ is called the *coefficient of viscosity* or the *dynamic* (or *absolute*) *viscosity* of the fluid, whose unit is kg/m·s, or equivalently, N·s/m^2 (or Pa·s where Pa is the pressure unit pascal).

The viscosity of a fluid is a measure of its "resistance to deformation." Viscosity is due to the internal frictional force that develops between different layers of fluids as they are forced to move relative to each other.

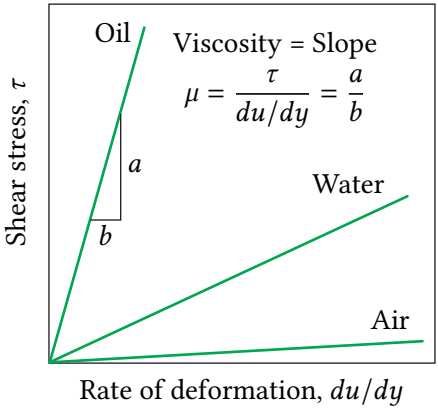
4.4.1 Absolute viscosity

In one-dimensional shear flow of Newtonian fluids, shear stress can be expressed by the linear relationship (*Newton's law of viscosity*)

$$\tau = \mu \frac{du}{dy} \qquad (\mathrm{N/m^2}).$$

The constant of proportionality μ is called the *coefficient of viscosity* or the *absolute viscosity* or the *dynamic viscosity* of the fluid, whose unit is kg/m·s, or equivalently, N·s/m^2 (or Pa·s where Pa is the pressure unit pascal).

A common viscosity unit is poise, which is equivalent to 0.1 Pa·s (or *centipoise*, which is one-hundredth of a poise). The viscosity of water at 20°C is 1.002 centipoise, and thus the unit centipoise serves as a useful reference. A plot of shear stress versus the rate of deformation (velocity gradient) for a Newtonian fluid is a straight line whose slope is the viscosity of the fluid, as shown in the figure. Note that viscosity is independent of the rate of deformation for Newtonian fluids. Since the rate of deformation is proportional to the strain rate, the plot reveals that viscosity is actually a coefficient in a stress–strain relationship.



The shear force acting on a Newtonian fluid layer (or, by Newton's third law, the force acting on the plate) is

$$F = \tau A = \mu A \frac{du}{dy}$$

where A is the contact area between the plate and the fluid. Then the force F required to move the upper plate at a constant speed of V while the lower plate remains stationary is

$$F = \mu A \frac{V}{l} \qquad \text{(N)}$$

This relation can alternately be used to calculate μ when the force F is measured. Therefore, the experimental setup just described can be used to measure the viscosity of fluids. Note that under identical conditions, the force F would be very different for different fluids.

Following table gives dynamic viscosity of some fluids at 1 atm and 20°C (unless otherwise stated).

Fluid	Dynamic Viscosity μ, N s/m^2 or kgm^{-1}s^{-1}
Glycerin: at	
−20°C	134.0
0°C	10.5
20°C	1.52
40°C	0.31
Mercury	0.0015
Ethyl alcohol	0.0012
Water	0.0010
Air	0.000018

4.4.2 Kinematic viscosity

In fluid mechanics and heat transfer, the ratio of dynamic viscosity to density appears frequently. For convenience, this ratio is given the name *kinematic viscosity, v* and is expressed as

$$v = \mu/\rho.$$

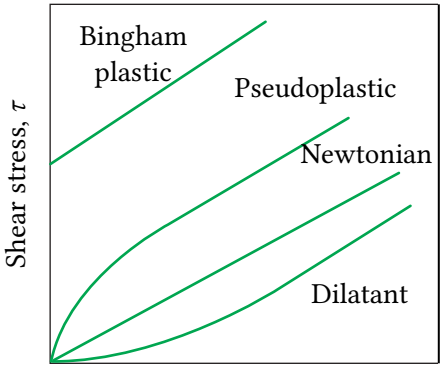
Two common units of kinematic viscosity are m^2/s and *stoke* (1 stoke = 1 cm^2/s = 0.0001 m^2/s).

In general, the viscosity of a fluid depends on both temperature and pressure, although the dependence on pressure is rather weak. For liquids, both the dynamic and kinematic viscosities are practically independent of pressure, and any small variation with pressure is usually disregarded, except at extremely high pressures. For gases, this is also the case for dynamic viscosity (at low to

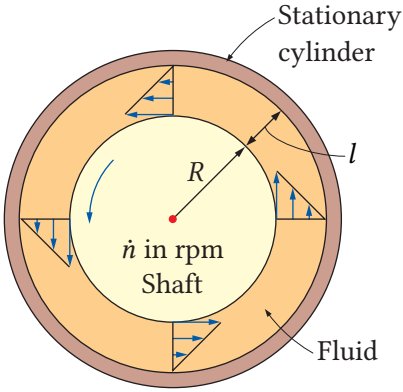
moderate pressures), but not for kinematic viscosity since the density of a gas is proportional to its pressure.

4.4.3 Non-Newtonian fluids

For non-Newtonian fluids, the relationship between shear stress and rate of deformation is not linear, as shown in the figure. The slope of the curve on the τ versus du/dy chart is referred to as the *apparent viscosity* of the fluid. Fluids for which the apparent viscosity increases with the rate of deformation (such as solutions with suspended starch or sand) are referred to as *dilatant* or *shear thickening fluids*, and those that exhibit the opposite behavior (the fluid becoming less viscous as it is sheared harder, such as some paints, polymer solutions, and fluids with suspended particles) are referred to as *pseudoplastic* or *shear thinning fluids*. Some materials such as toothpaste can resist a finite shear stress and thus behave as a solid, but deform continuously when the shear stress exceeds the yield stress and behave as a fluid. Such materials are referred to as Bingham plastics after Eugene C Bingham (1878–1945), who did pioneering work on fluid viscosity for the U.S. National Bureau of Standards in the early twentieth century.



4.4.4 Concentric cylinders rotational viscometer



Rotational viscometers use the idea that the torque required to rotate an object in a fluid is a function of the viscosity of that fluid. They measure the torque required to rotate a disk or bob in a fluid at a known speed.

Consider a fluid layer of thickness l, within a small gap between two concentric cylinders, such as the thin layer of oil in a journal bearing. The gap between the cylinders can be modeled as two parallel flat plates separated by the fluid. Noting that torque is $T = FR$ (force times the moment arm, which is the radius R of the inner cylinder in this case), the tangential velocity is $V = \omega R$ (angular velocity times the radius), and taking the wetted surface area of the inner cylinder to be $A = 2\pi RL$, where L is the length of the cylinder, by disregarding the shear stress acting on the two ends of the inner cylinder, torque can be expressed as

$$T = FR = \mu A\frac{V}{l}R = \mu 2\pi RL\frac{\omega R}{l}R = \mu\frac{2\pi R^3\omega L}{l} = \mu\frac{4\pi^2R^3\dot{n}L}{l}$$

where L is the length of the cylinder and $\dot{n}$ is the number of revolutions per unit time, which is usually expressed in rpm (revolutions per minute). Note that the angular distance traveled during one rotation is 2π rad, and thus the relation between the angular velocity in rad/min and the rpm is $\omega = 2\pi\dot{n}$. The above equation can be used to calculate the viscosity of a fluid by measuring torque

at a specified angular velocity. Therefore, two concentric cylinders can be used as a viscometer, a device that measures viscosity.

Example 1. The viscosity of a fluid is to be measured by a viscometer constructed of two 40 cm long concentric cylinders. The outer diameter of the inner cylinder is 12 cm, and the gap between the two cylinders is 0.15 cm. The inner cylinder is rotated at 300 rpm, and the torque is measured to be 1.8 Nm. Determine the viscosity of the fluid.

Solution. *Assumptions:* 1) The inner cylinder is completely submerged in the fluid. 2) The viscous effects on the two ends of the inner cylinder are negligible.

The velocity profile is linear only when the curvature effects are negligible, and the velocity profile can be approximated as being linear in this case since $l/R = 0.15\text{cm}/6\text{cm} = 0.025 \ll 1$. Solving above equation for viscosity and substituting the given values, the viscosity of the fluid is determined to be

$$\mu = \frac{Tl}{4\pi^2 R^3 \dot{n} L} = \frac{1.8\,\text{Nm} \times 0.0015\,\text{m}}{4\pi^2 \times (0.06\,\text{m})^3 \times \frac{300}{60\text{s}} \times 0.4\,\text{m}} = 0.158\,\text{Nsm}^{-2}$$

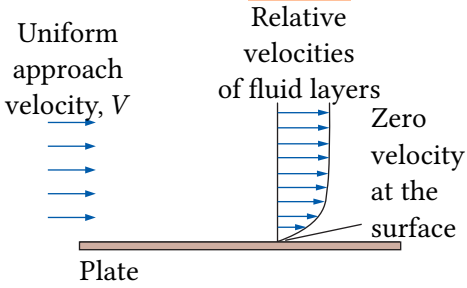
Viscosity is a strong function of temperature, and a viscosity value without a corresponding temperature is of little usefulness. Therefore, the temperature of the fluid should have also been measured during this experiment, and reported with this calculation. □

4.5 The no-slip condition

Fluid flow is often confined by solid surfaces, and it is important to understand how the presence of solid surfaces affects fluid flow. We know that water in a river cannot flow through large rocks, and must go around them. That is, the water velocity normal to the rock surface must be zero, and water approaching the surface normally comes to a complete stop at the surface. What is not as obvious is that water approaching the rock at any angle also comes to a complete stop at the rock surface, and thus the tangential velocity of water at the surface is also zero.

Consider the flow of a fluid in a stationary pipe or over a solid surface that is non-porous (i.e., impermeable to the fluid). All experimental observations

indicate that a fluid in motion comes to a complete stop at the surface and assumes a zero velocity relative to the surface. That is, a fluid in direct contact with a solid "sticks" to the surface, and there is no slip. This is known as the *no-slip condition*. The fluid property responsible for the no-slip condition and the development of the boundary layer is *viscosity*.

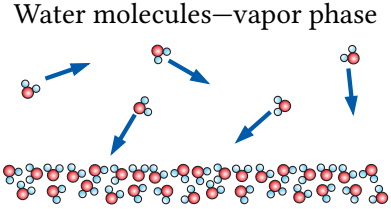


A consequence of the no-slip condition is that all velocity profiles must have zero values with respect to the surface at the points of contact between a fluid and a solid surface. Therefore, the no-slip condition is responsible for the development of the velocity profile. The flow region adjacent to the wall in which the viscous effects (and thus the velocity gradients) are significant is called the *boundary layer*. Another consequence of the no-slip condition is the *surface drag*, or *skin friction drag*, which is the force a fluid exerts on a surface in the flow direction.

4.6 Vapour pressure and cavitation

It is well-established that temperature and pressure are dependent properties for pure substances during phase-change processes, and there is one-to-one correspondence between temperature and pressure. At a given pressure, the temperature at which a pure substance changes phase is called the *saturation temperature* T_{sat}. Likewise, at a given temperature, the pressure at which a pure substance changes phase is called the *saturation pressure* P_{sat}. At an absolute pressure of 1 standard atmosphere (1 atm or 101.325 kPa), for example, the saturation temperature of water is 100°C. Conversely, at a temperature of 100°C, the saturation pressure of water is 1 atm.

The *vapor pressure* P_v of a pure substance is defined as *the pressure exerted by its vapor in phase equilibrium with its liquid at a given temperature*. P_v is a property of the pure substance, and turns out to be identical to the saturation pressure P_{sat} of the liquid ($P_v = P_{sat}$). We must be careful not to confuse vapor pressure with *partial pressure*. Partial pressure is defined as *the pressure of a gas or vapor in a mixture with other gases*. For example, atmospheric air is a mixture of dry air and water vapor, and atmospheric pressure is the sum of the partial pressure of dry air and the partial pressure of water vapor.



The reason for our interest in vapor pressure is the possibility of the liquid pressure in liquid-flow systems dropping below the vapor pressure at some locations, and the resulting unplanned vaporization. For example, water at 10°C may vaporize and form bubbles at locations (such as the tip regions of impellers or suction sides of pumps) where the pressure drops below 1.23 kPa. The vapor bubbles (called *cavitation bubbles* since they form "cavities" in the liquid) collapse as they are swept away from the low-pressure regions, generating highly destructive, extremely high-pressure waves. This phenomenon, which is a common cause for drop in performance and even the erosion of impeller blades, is called *cavitation*, and it is an important consideration in the design of hydraulic turbines and pumps.

Cavitation must be avoided (or at least minimized) in most flow systems since it reduces performance, generates annoying vibrations and noise, and causes damage to equipment. We note that some flow systems use cavitation to their *advantage*, e.g., high-speed "supercavitating" torpedoes. The pressure spikes resulting from the large number of bubbles collapsing near a solid surface over a long period of time may cause erosion, surface pitting, fatigue failure,

and the eventual destruction of the components or machinery. The presence of cavitation in a flow system can be sensed by its characteristic tumbling sound.



This is a photo of Cavitation damage on a 16-mm by 23-mm aluminum sample tested at 60 m/s for 2.5 hours. The sample was located at the cavity collapse region downstream of a cavity generator specifically designed to produce high damage potential.

Cavitation is the rupture of a liquid, or of a fluid–solid interface, caused by a reduction of the local static pressure produced by the dynamic action of the fluid in the interior and/or boundaries of a liquid system. The rupture is the formation of a visible bubble. Liquids, such as water, contain many microscopic voids that act as *cavitation nuclei*. Cavitation occurs when these nuclei grow to a significant, visible size. Although boiling is also the formation of voids in a liquid, we usually separate this phenomenon from cavitation because it is caused by an increase in temperature, rather than by a reduction in pressure. Cavitation can be used in beneficial ways, such as in ultrasonic cleaners, etchers, and cutters. But more often than not, cavitation is to be avoided in fluid flow applications because it spoils hydrodynamic performance, it causes extremely loud noise and high vibration levels, and it damages (erodes) the surfaces that support it. When cavitation bubbles enter regions of high pressure and collapse, the underwater shock waves sometimes create minute amounts of light. This phenomenon is called *sonoluminescence*.

4.7 Bulk Modulus and compressibility

The volume (or density) of a fluid changes with a change in its temperature or pressure. Fluids usually expand as they are heated or depressurized and contract as they are cooled or pressurized. But the amount of volume change is different for different fluids, and we need to define properties that relate volume changes to the changes in pressure and temperature. Two such properties are the bulk modulus of elasticity κ and the coefficient of volume expansion β.

It is a common observation that a fluid contracts when more pressure is applied on it and expands when the pressure acting on it is reduced. That is, fluids act like elastic solids with respect to pressure. Therefore, in an analogous manner to Young's modulus of elasticity for solids, it is appropriate to define a *coefficient of compressibility* κ (also called the *bulk modulus of compressibility* or *bulk modulus of elasticity*) for fluids as

$$\kappa = -V_\rho \left(\frac{\partial P}{\partial V_\rho}\right)_T = \rho \left(\frac{\partial P}{\partial \rho}\right)_T \qquad \text{(in Pa)}.$$

Where V_ρ is the specific volume ($V_\rho = \frac{V_m}{m} = \frac{1}{\rho}$) and ρ is the density of the fluid, P is the pressure applied on the fluid.

The bulk modulus κ can also be expressed approximately in terms of finite changes as

$$\kappa = \frac{\text{stress}}{\text{strain}} \approx \frac{-\Delta P}{\Delta V_m / V_m} = \frac{-\Delta P}{\Delta V_\rho / V_\rho} = \frac{\Delta P}{\Delta \rho / \rho} \qquad (T = \text{constant}).$$

Noting that $\Delta V_\rho / V_\rho$ or $\Delta\rho/\rho$ is dimensionless, κ must have the dimension of pressure (Pa or psi). Also, the coefficient of compressibility represents the change in pressure corresponding to a fractional change in volume or density of the fluid while the temperature remains constant. Then it follows that the coefficient of compressibility of a truly incompressible substance (V_ρ = constant) is infinity.

A large value of κ indicates that a large change in pressure is needed to cause a small fractional change in volume, and thus a fluid with a large k is essentially incompressible. This is typical for liquids, and explains why liquids

are usually considered to be *incompressible*. For example, the pressure of water at normal atmospheric conditions must be raised to 210 atm to compress it 1 percent, corresponding to a coefficient of compressibility value of $\kappa = 21,000$ atm.

Note that volume and pressure are inversely proportional (volume decreases as pressure is increased and thus $\partial P/\partial V_\rho$ is a negative quantity), and the negative sign in the above definition ensures that κ is a positive quantity. Also, differentiating $\rho = 1/V_\rho$ gives $d\rho = -dV_\rho/V_\rho^2$, which can be rearranged as

$$\frac{d\rho}{\rho} = -\frac{dV_\rho}{V_\rho}$$

That is, the fractional changes in the specific volume and the density of a fluid are equal in magnitude but opposite in sign.

The inverse of the coefficient of compressibility is called the *isothermal compressibility* α and is expressed as

$$\alpha = \frac{1}{\kappa} = -\frac{1}{V_\rho}\left(\frac{\partial V_\rho}{\partial P}\right)_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial P}\right)_T \qquad (\text{in Pa}^{-1}).$$

The isothermal compressibility of a fluid represents the fractional change in volume or density corresponding to a unit change in pressure.

Example 2. Show that the bulk modulus of an ideal gas is equal to its absolute pressure.

Solution. For an ideal gas,

$$PV_m = mRT$$

or,

$$P = \frac{m}{V_m}RT = \rho RT$$

where P is the pressure, V_m is the volume, m is the mass, R is the gas constant, T is the temperature, M is the molar mass (also called molecular weight) of the

gas molecules, $m = nM$ is the total mass of the gas and ρ is its density. The bulk modulus is given by,

$$\kappa_{\text{ideal gas}} = \rho \left(\frac{\partial P}{\partial \rho}\right)_T = \rho \left(\frac{\partial(\rho RT)}{\partial \rho}\right)_T = \rho\,(RT) = P$$

Therefore, the coefficient of compressibility of an ideal gas is equal to its absolute pressure, and the coefficient of compressibility of the gas increases with increasing pressure. Substituting $\kappa = P$ into the definition of the coefficient of compressibility and rearranging gives

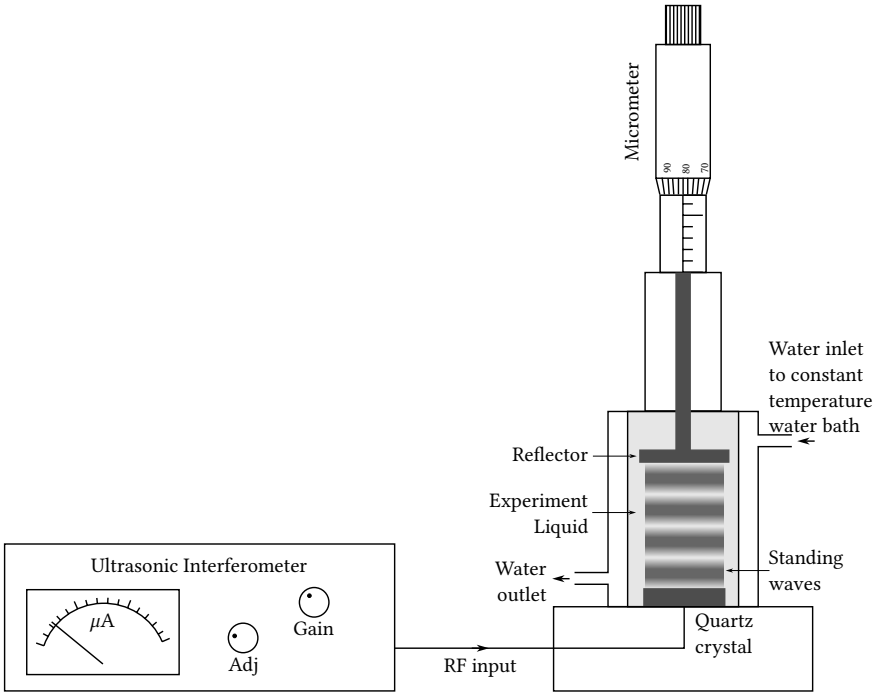
$$\text{For ideal gas:} \qquad \frac{\Delta\rho}{\rho} = \frac{\Delta P}{P} \qquad (T = \text{constant})$$

Therefore, the percent increase of density of an ideal gas during isothermal compression is equal to the percent increase in pressure. □

4.8 Ultrasonic interferometer

Ultrasound is sound waves with frequencies higher than the upper audible limit of human hearing. Ultrasound is not different from "normal" (audible) sound in its physical properties, except that humans cannot hear it. This limit varies from person to person and is approximately 20 kilohertz (20,000 hertz) in healthy young adults. Ultrasound devices operate with frequencies from 20 kHz up to several gigahertz.

Ultrasonic interferometer is a device which can measure wavelength of ultrasound in a liquid medium with a high degree of accuracy. From the measured wavelength and using frequency of the applied alternating voltage to produce the sound waves, velocity of the ultrasonic waves in the medium is determined.



4.8.1 Construction

In an ultrasonic interferometer, the ultrasonic waves are produced by the piezoelectric methods. Ultrasonic transducers convert AC into ultrasound, as well as the reverse. They typically refers to piezoelectric transducers. Piezoelectric crystals change the size and shape when a voltage is applied. AC voltage makes them oscillate at the same frequency as that of the applied alternating voltage and produce ultrasonic waves. Since piezoelectric materials generate a voltage when force is applied to them, they can also work as ultrasonic detectors. Some systems use separate transmitters and receivers, while others combine both functions into a single piezoelectric transceiver.

At a fixed frequency variable path interferometer, the wavelength of the sound in an experimental liquid medium is measured, and from this one can

calculate its velocity through that medium. Ultrasonic waves of known frequency are produced by a quartz crystal which is fixed at the bottom of the cell. There is a movable metallic plate parallel to the quartz plate, which reflects the waves. The waves interfere with their reflections, and if the separation between the plates is exactly an integer multiple of half wave length of sound, standing waves are produced in the liquid medium. Under these circumstances, acoustic resonance occurs. The resonant waves are maximum in amplitude, causing a corresponding maximum in the anode current of the piezoelectric generator. The ultrasonic interferometer consists of the following mainly two parts:

1. **The high frequency generator:** The high frequency generator is designed to excite the quartz crystal fixed at the bottom of the measuring cell at its resonant frequency to generate ultrasonic waves in the experimental liquid filled in the "measuring cell". A micro-ammmeter is proveided to observe the changes in current flowing to the quartz crystal.

   For initial adjustment two knobs are provided on high frequency generator, one is marked with "ADJ" to adjust the position of the needle on the ammeter and the knob marked "GAIN" is used to increase the sensitivity of the instrument for greater deflection, if desired. The ammeter is used to notice the number of maximum deflection while micrometer is moved up or down in liquid.

2. **The measuring cell:** The measuring cell is specially designed for maintaining the temperature of the liquid constant during the experiment. The cell consists of a double walled brass cell with chromium plated surfaces having a capacity of 10 ml. The double wall allows water circulation around the experimental liquid to maintain it at a known constant temperature. The micrometer scale is marked in units of 0.01 mm and has an overall length of 25 mm. A fine micrometer screw (LC = 0.001 mm) is provided at the top, which can lower or raise the reflector plate in the liquid within the cell through a known distance. It has a quartz crystal fixed at its bottom.

4.8.2 Working Principle

The principle used in the measurement of velocity c is based on the accurate determination of the wavelength λ in the medium. Ultrasonic waves of known frequency f are produced by quartz crystal fixed at the bottom of the cell. These waves are reflected by a movable metallic plate kept parallel to the quartz crystal. If the separation between these two plates is exactly a whole multiple of the wavelength, standing waves are formed in the medium. This acoustic resonance gives rise to an electrical reaction on the generator driving the quartz crystal and anode current of the generator become a maximum. If the distance is now increased or decreased and the variation is exactly one half wavelengths $\lambda/2$ or integer multiple of it, anode current becomes maximum again. From the knowledge of wavelength the velocity can be obtained by the relation:

$$\text{Velocity} = \text{Wavelength} \times \text{Frequency}$$

$$c = \lambda f$$

4.8.3 Procedure

The cap of cell is unscrewed and lifted it away from the double walled construction of the cell. In the middle position of it experimental liquid is poured and the cap is screwed. Excess liquid overflowing from the cell is wiped out. The cell is inserted in the heavy base socket and is clamped with the help of a screw provided on its side. The high frequency generator is connected with the cell by a co-axial cable provided with the instrument. Frequency selector knob should be positioned at the desired frequency. The micrometer is slowly moved in either clockwise or anticlockwise direction till the anode current on the ammeter shows a maximum or minimum. The micrometer reading corresponding to the maximum or minimum (which is sharper) in micro-ammeter is noted. About 50 readings of consecutive maximum or minimum are noted and tabulated. The average of all differences gives $\lambda/2$. Once the wavelength λ is known the velocity of ultrasound c in the liquid can be calculated with the help of the above relation.

4.8.4 Calculation

The velocity of ultrasound is determined principally by the compressibility of the material of the medium. For a medium with high compressibility, the velocity will be less. Adiabatic compressibility of a fluid is a measure of the relative volume change of the fluid as a response to a pressure change. Compressibility is the reciprocal of bulk modulus. The *adiabatic compressibility* of the material of the sample can be calculated using the equation,

$$\alpha_{\text{ad}} = \frac{1}{\kappa} = \frac{\gamma}{\rho c^2}$$

Where ρ is the density of the material of the medium, c is the velocity of the sound wave through that medium and $\gamma = c_p/c_v$ is the *specific heat ratio* of the fluid. For liquids $\gamma = 1$, hence

$$\alpha_{\text{ad}} = \frac{1}{\rho c^2}.$$

Example 3. An ultrasonic interferometer is operated with alternating current of 2×10^6 Hz. With a liquid of density 996.45 kg/m^3 placed in its measuring cell, two consecutive micrometer readings for minimum current are found to be 1.100 mm and 1.349 mm. Determine adiabatic compressibility of the liquid in the cell.

Solution. Here $d = 1.35\,\text{mm} - 1.10\,\text{mm} = 0.25\,\text{mm}$ and by the principle of ultrasonic interferometer,

$$d = \lambda/2$$

Hence,

$$\lambda = 2d = 2 \times 0.25\,\text{mm} = 0.5\,\text{mm}$$

Therefore velocity of the ultrasound in the medium is,

$$c = \lambda f = 0.5 \times 10^{-3}\,\text{m} \times 2 \times 10^6\,\text{Hz} = 1000\,\text{m/s}$$

Adiabatic compressibility is,

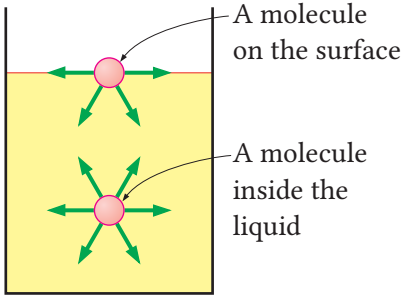
$$\alpha_{ad} = \frac{1}{\rho c^2} = \frac{1}{996.45\,\text{kg m}^{-3} \times (1000)^2 \text{m}^2\text{s}^{-2}}$$
$$= 1.00356 \times 10^{-9}\text{m}^2\text{N}^{-1} = 1.00356 \times 10^{-9}\text{Pa}^{-1}$$

□

4.9 Surface tension

It is often observed that a drop of mercury forms a near-perfect sphere and can be rolled just like a steel ball over a smooth surface; water droplets from rain or dew hang from branches or leaves of trees; water dripping from a leaky tap falls as nearly spherical droplets; a soap bubble released into the air forms a nearly spherical shape; and water beads up into small drops on flower petals.

In these and other observances, liquid droplets behave like small balloons filled with the liquid, and the surface of the liquid acts like a stretched elastic membrane under tension. The pulling force that causes this tension acts parallel to the surface and is due to the attractive forces between the molecules of the liquid. The magnitude of this force per unit length is called *surface tension* or *coefficient of surface tension* σ_s and is usually expressed in the unit N/m. This effect is also called *surface energy* (per unit area) and is expressed in the equivalent unit of Nm/m^2 or J/m^2. In this case, σ_s represents the stretching work that needs to be done to increase the surface area of the liquid by a unit amount.



To visualize how surface tension arises, we present a microscopic view in above figure by considering two liquid molecules, one at the surface and one deep within the liquid body. The attractive forces applied on the interior molecule by the surrounding molecules balance each other because of symmetry. But the attractive forces acting on the surface molecule are not symmetric, and the attractive forces applied by the gas molecules above are usually very small. Therefore, there is a net attractive force acting on the molecule at the surface of the liquid, which tends to pull the molecules on the surface toward the interior of the liquid. This force is balanced by the repulsive forces from the molecules below the surface that are trying to be compressed. The result is that the liquid minimizes its surface area. This is the reason for the tendency of liquid droplets to attain a spherical shape, which has the minimum surface area for a given volume.

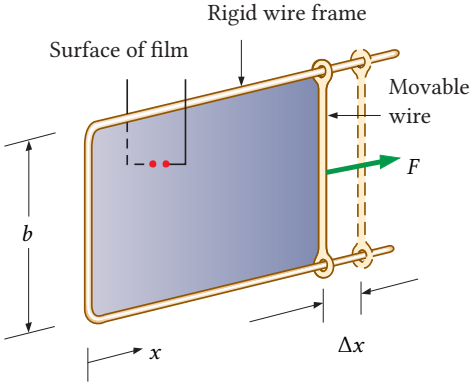
You also may have observed, with amusement, that some insects can land on water or even walk on water and that small steel needles can float on water. These phenomena are made possible by surface tension which balances the weights of these objects.

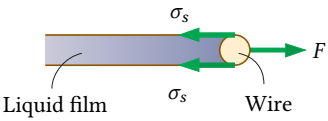
4.9.1 Liquid film suspended on a U-shaped wire frame

To understand the surface tension effect better, consider a liquid film (such as the film of a soap bubble) suspended on a U-shaped wire frame with a movable side as shown in the figure. Normally, the liquid film tends to pull the movable wire inward in order to minimize its surface area. A force F needs to be applied on the movable wire in the opposite direction to balance this pulling effect. Both sides of the thin film are surfaces exposed to air, and thus the length along which the surface tension acts in this case is $2b$. Then a force balance on the movable wire gives $F = 2b\sigma_s$, and thus the surface tension can be expressed as

$$\sigma_s = \frac{F}{2b}$$

Note that for $b = 0.5$ m, the measured force F (in N) is simply the surface tension in N/m. An apparatus of this kind with sufficient precision can be used to measure the surface tension of various liquids.





In the U-shaped wire frame apparatus, the movable wire is pulled to stretch the film and increase its surface area. When the movable wire is pulled a distance Δx, the surface area increases by $\Delta A = 2b\,\Delta x$, and the work W done during this stretching process is

$$W = \text{Force} \times \text{Distance} = F\,\Delta x = 2b\sigma_s\,\Delta x = \sigma_s\,\Delta A$$

where we have assumed that the force remains constant over the small distance. This result can also be interpreted as *the surface energy of the film is increased by an amount* $\sigma_s\,\Delta A$ *during this stretching process*, which is consistent with the alternative interpretation of σ_s as surface energy per unit area. This is similar to a rubber band having more potential (elastic) energy after it is stretched further. In the case of liquid film, the work is used to move liquid molecules from the interior parts to the surface against the attraction forces of other molecules. Therefore, surface tension also can be defined as the *work done per unit increase in the surface area of the liquid.*

4.9.2 Variations in surface tension

The surface tension varies greatly from substance to substance, and with temperature for a given substance, as shown in the table. At 20°C, for example, the surface tension is 0.073 N/m for water and 0.440 N/m for mercury surrounded by atmospheric air. The surface tension of mercury is large enough that mercury droplets form nearly spherical balls that can be rolled like a solid ball on

a smooth surface. The surface tension of a liquid, in general, decreases with temperature and becomes zero at the critical point (and thus there is no distinct liquid–vapor interface at temperatures above the critical point). The effect of pressure on surface tension is usually negligible.

Fluid	Surface Tension σ_s, in N/m
Water: at	
0°C	0.076
20°C	0.073
100°C	0.059
300°C	0.014
Glycerin	0.063
Mercury	0.440
Ethyl alcohol	0.023
Soap solution	0.025
Kerosene	0.028

The above table gives surface tension of some fluids in air at 1 atm and 20°C (unless otherwise stated).

The surface tension of a substance can be changed considerably by *impurities*. Therefore, certain chemicals, called *surfactants*, can be added to a liquid to decrease its surface tension. For example, soaps and detergents lower the surface tension of water and enable it to penetrate the small openings between fibers for more effective washing. But this also means that devices whose operation depends on surface tension (such as heat pipes) can be destroyed by the presence of impurities due to poor workmanship.

4.9.3 Excess pressure inside a droplet and a soap bubble

Surface tension determines the size of the liquid droplets that form, and so a droplet that keeps growing by the addition of more mass breaks down when the surface tension can no longer hold it together. This is like a balloon that bursts while being inflated when the pressure inside rises above the strength of the balloon material.

A curved interface indicates a pressure difference (or "pressure jump") across the interface with pressure being higher on the concave side. Consider, for example, a droplet of liquid in air, an air (or other gas) bubble in water, or a soap bubble in air. The excess pressure ΔP above atmospheric pressure can be determined by considering a free-body diagram of half the droplet or bubble (see the figure). Noting that surface tension acts along the circumference and the pressure acts on the area, horizontal force balances for the droplet or air bubble and the soap bubble give:

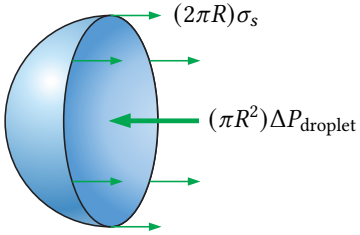
For droplet or air bubble:

$$(2\pi R)\sigma_s = (\pi R^2)\,\Delta P_{\text{droplet}} \qquad \Longrightarrow \quad \Delta P_{\text{droplet}} = P_i - P_o = \frac{2\sigma_s}{R}$$

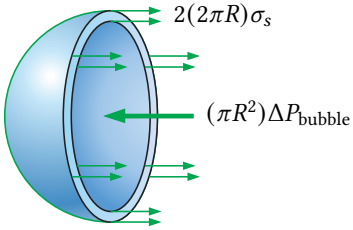
For soap bubble:

$$2(2\pi R)\sigma_s = (\pi R^2)\,\Delta P_{\text{bubble}} \qquad \Longrightarrow \quad \Delta P_{\text{bubble}} = P_i - P_o = \frac{4\sigma_s}{R}$$

where P_i and P_o are the pressures inside and outside the droplet or bubble, respectively. When the droplet or bubble is in the atmosphere, P_o is simply atmospheric pressure. The extra factor of 2 in the force balance for the soap bubble is due to the existence of a soap film with *two* surfaces (inner and outer surfaces) and thus two circumferences in the cross section.



(a) Half of a droplet or air bubble



(b) Half of a soap bubble

The excess pressure in a droplet of liquid in a gas (or a bubble of gas in a liquid) can also be determined by considering a differential increase in the radius

of the droplet due to the addition of a differential amount of mass and interpreting the surface tension as the increase in the surface energy per unit area. Then the increase in the surface energy of the droplet during this differential expansion process becomes

$$\delta W_{\text{surface}} = \sigma_s dA = \sigma_s d(4\pi R^2) = 8\pi R \sigma_s dR$$

The expansion work done during this differential process is determined by multiplying the force by distance to obtain

$$\delta W_{\text{expansion}} = \text{Force} \cdot \text{Distance} = F\,dR = (\Delta P A)\,dR = 4\pi R^2 \Delta P\,dR$$

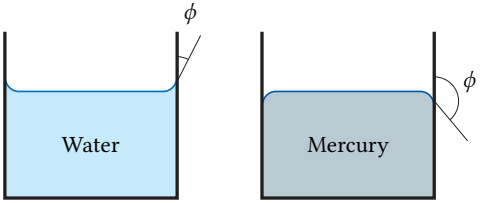
Equating the two expressions above gives

$$\Delta P_{\text{droplet}} = \frac{2\sigma_s}{R},$$

which is the same relation obtained before. Note that the excess pressure in a droplet or bubble is inversely proportional to the radius.

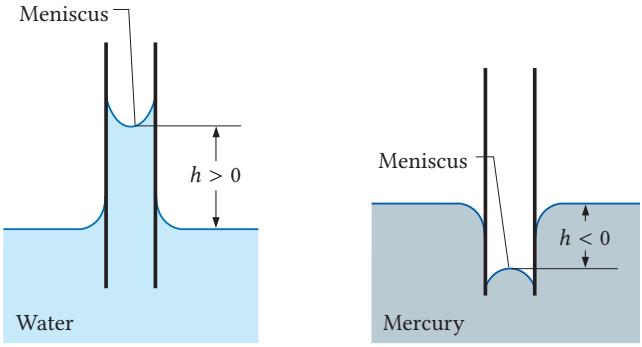
4.9.4 Capillarity

Another interesting consequence of surface tension is the *capillary effect*, which is the rise or fall of a liquid in a small-diameter tube inserted into the liquid. Such narrow tubes or confined flow channels are called *capillaries*. The rise of kerosene through a cotton wick inserted into the reservoir of a kerosene lamp is due to this effect. The capillary effect is also partially responsible for the rise of water to the top of tall trees. The curved free surface of a liquid in a capillary tube is called the *meniscus*.



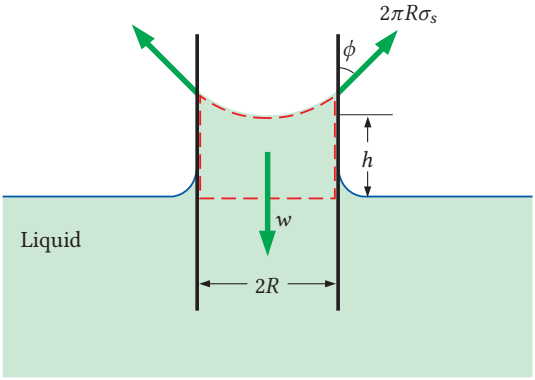
(a) Wetting fluid (b) Nonwetting fluid

It is commonly observed that water in a glass container curves up slightly at the edges where it touches the glass surface; but the opposite occurs for mercury: it curves down at the edges (See the above figure). This effect is usually expressed by saying that water *wets* the glass (by sticking to it) while mercury does not. The strength of the capillary effect is quantified by the *contact* (or *wetting*) angle ϕ, defined as *the angle that the tangent to the liquid surface makes with the solid surface at the point of contact*. The surface tension force acts along this tangent line toward the solid surface. A liquid is said to wet the surface when $\phi < 90°$ and not to wet the surface when $\phi > 90°$. In atmospheric air, the contact angle of water (and most other organic liquids) with glass is nearly zero, $\phi \approx 0°$. Therefore, the surface tension force acts upward on water in a glass tube along the circumference, tending to pull the water up. As a result, water rises in the tube until the weight of the liquid in the tube above the liquid level of the reservoir balances the surface tension force. The contact angle is 130° for mercury–glass and 26° for kerosene–glass in air. Note that the contact angle, in general, is different in different environments (such as another gas or liquid in place of air).



The phenomenon of the capillary effect can be explained microscopically by considering *cohesive forces* (the forces between like molecules, such as water and water) and *adhesive forces* (the forces between unlike molecules, such as water and glass). The liquid molecules at the solid–liquid interface are subjected to both cohesive forces by other liquid molecules and adhesive forces by the molecules of the solid. The relative magnitudes of these forces determine

whether a liquid wets a solid surface or not. Obviously, the water molecules are more strongly attracted to the glass molecules than they are to other water molecules, and thus water tends to rise along the glass surface. The opposite occurs for mercury, which causes the liquid surface near the glass wall to be suppressed (see the above figure).



The magnitude of the capillary rise in a circular tube can be determined from a force balance on the cylindrical liquid column of height h in the tube (see above figure). The bottom of the liquid column is at the same level as the free surface of the reservoir, and thus the pressure there must be atmospheric pressure. This balances the atmospheric pressure acting at the top surface of the liquid column, and thus these two effects cancel each other. The weight of the liquid column is approximately

$$W = mg = \rho V_m g = \rho g(\pi R^2 h)$$

Equating the vertical component of the surface tension force to the weight gives

$$W = F_{\text{surface}}$$

$$\Longrightarrow \quad \rho g(\pi R^2 h) = 2\pi R\sigma_s \cos\phi$$

Solving for h gives the capillary rise to be

$$h = \frac{2\sigma_s}{\rho g R}\cos\phi \qquad (R = \text{constant})$$

This relation is also valid for non-wetting liquids (such as mercury in glass) and gives the capillary drop. In this case $\phi > 90^\circ$ and thus $\cos\phi < 0$, which makes h negative. Therefore, a negative value of capillary rise corresponds to a capillary drop.

Note that the capillary rise is inversely proportional to the radius of the tube. Therefore, the thinner the tube is, the greater the rise (or fall) of the liquid in the tube. In practice, the capillary effect for water is usually negligible in tubes whose diameter is greater than 1 cm. When pressure measurements are made using manometers and barometers, it is important to use sufficiently large tubes to minimize the capillary effect. The capillary rise is also inversely proportional to the density of the liquid, as expected. Therefore, in general, lighter liquids experience greater capillary rises. Finally, it should be kept in mind that the above equation is derived for constant-diameter tubes and should not be used for tubes of variable cross section.

Example 4. A 0.6 mm diameter glass tube is inserted into water at 20°C in a cup. Determine the capillary rise of water in the tube. Given the surface tension of water at 20°C is 0.073 N/m.

Solution. *Assumptions:* 1) There are no impurities in the water and no contamination on the surfaces of the glass tube. 2) The experiment is conducted in atmospheric air.

Properties: The surface tension of water at 20°C is 0.073 N/m. The contact angle of water with glass is approximately 0°. We take the density of liquid water to be 1000 kg/m^3.

Analysis: The capillary rise is determined directly from,

$$h = \frac{2\sigma_s}{\rho g R}\cos\phi = \frac{2(0.073\,\text{N/m})}{(1000\,\text{kg/m}^3)(9.81\,\text{m/s}^2)(0.3\times10^{-3}\,\text{m})}(\cos 0^\circ)\left(\frac{1\text{kg}\cdot\text{m/s}^2}{1\,\text{N}}\right)$$
$$= 0.050\,\text{m} = 5.0\,\text{cm}$$

Therefore, water rises in the tube 5 cm above the liquid level in the cup. □

5 Types of Fluid Flows

There is a wide variety of fluid flow problems encountered in practice, and it is usually convenient to classify them on the basis of some common characteristics to make it feasible to study them in groups. There are many ways to classify fluid flow problems, and here we present some general categories.

1. **Viscous versus Inviscid Regions of Flow:** When two fluid layers move relative to each other, a friction force develops between them and the slower layer tries to slow down the faster layer. This internal resistance to flow is quantified by the fluid property *viscosity*, which is a measure of internal stickiness of the fluid. Viscosity is caused by cohesive forces between the molecules in liquids and by molecular collisions in gases. There is no fluid with zero viscosity, and thus all fluid flows involve viscous effects to some degree. Flows in which the frictional effects are significant are called *viscous flows*. However, in many flows of practical interest, there are regions (typically regions not close to solid surfaces) where viscous forces are negligibly small compared to inertial or pressure forces. Neglecting the viscous terms in such *inviscid flow regions* greatly simplifies the analysis without much loss in accuracy.

2. **Internal versus External Flow:** A fluid flow is classified as being internal or external, depending on whether the fluid flows in a confined space or over a surface. The flow of an unbounded fluid over a surface such as a plate, a wire, or a pipe is *external flow*. The flow in a pipe or duct is *internal flow* if the fluid is completely bounded by solid surfaces. Water flow in a pipe, for example, is internal flow, and airflow over a ball or over an exposed pipe during a windy day is external flow. The flow of liquids in a duct is called *open-channel flow* if the duct is only partially filled with the liquid and there is a free surface. The flows of water in rivers and irrigation ditches are examples of such flows.

   Internal flows are dominated by the influence of viscosity throughout the flow field. In external flows the viscous effects are limited to boundary layers near solid surfaces and to wake regions downstream of bodies.

3. **Compressible versus Incompressible Flow**: A flow is classified as being *compressible* or *incompressible*, depending on the level of variation of density during flow. Incompressibility is an approximation, in which the flow is said to be *incompressible* if the density remains nearly constant throughout. Therefore, the volume of every portion of fluid remains unchanged over the course of its motion when the flow is approximated as incompressible.

   The densities of liquids are essentially constant, and thus the flow of liquids is typically incompressible. Therefore, liquids are usually referred to as *incompressible substances*. A pressure of 210 atm, for example, causes the density of liquid water at 1 atm to change by just 1 percent. Gases, on the other hand, are highly compressible. A pressure change of just 0.01 atm, for example, causes a change of 1 percent in the density of atmospheric air. Small density changes of liquids corresponding to large pressure changes can still have important consequences. The irritating "water hammer" in a water pipe, for example, is caused by the vibrations of the pipe generated by the reflection of pressure waves following the sudden closing of the valves.

4. **Steady versus Unsteady Flow**: The term *steady* implies *no change of properties, velocity, temperature, etc., at a point with time*. The opposite of steady is *unsteady*. The term *uniform* implies *no change with location* over a specified region. These meanings are consistent with their everyday use.

5. **One-, Two-, and Three-Dimensional Flows**: A flow field is best characterized by its velocity distribution, and thus a flow is said to be one-, two-, or three-dimensional if the flow velocity varies in one, two, or three primary dimensions, respectively. A typical fluid flow involves a three-dimensional geometry, and the velocity may vary in all three (x, y, z) in rectangular dimensions, rendering the flow three-dimensional $\vec{V}(x, y, z)$. However, the variation of velocity in certain directions can be small relative to the variation in other directions and can be ignored with negligible error. In such cases, the flow can be modeled conveniently as being one-

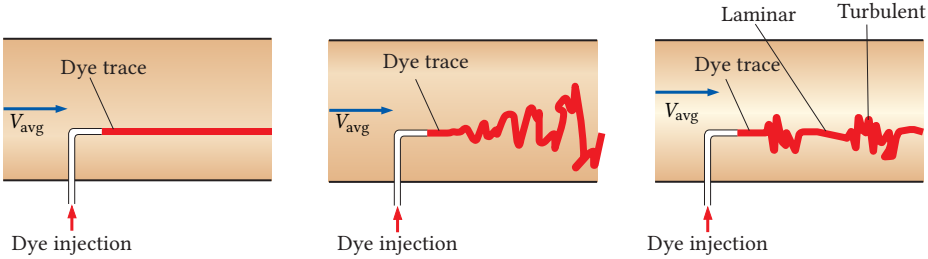
or two-dimensional, which is easier to analyze.

6. **Laminar and Turbulent Flows:** Discussed in next subsection.

5.1 Laminar and Turbulent Flows

Some flows are smooth and orderly while others are rather chaotic. The highly ordered fluid motion characterized by smooth layers of fluid is called *laminar*. The word *laminar* comes from the movement of adjacent fluid particles together in "laminae." The flow of high-viscosity fluids such as oils at low velocities is typically laminar. The highly disordered fluid motion that typically occurs at high velocities and is characterized by velocity fluctuations is called *turbulent*. The flow of low-viscosity fluids such as air at high velocities is typically turbulent. A flow that alternates between being laminar and turbulent is called *transitional*. The experiments conducted by Osborne Reynolds in the 1880s resulted in the establishment of the dimensionless *Reynolds number*, Re, as the key parameter for the determination of the flow regime in pipes.

You can notice that the smoke from a candle rises in a smooth plume for the first few centimeters and then starts fluctuating randomly in all directions as it continues its rise. Other plumes behave similarly. Likewise, a careful inspection of flow in a pipe reveals that the fluid flow is streamlined at low velocities but turns chaotic as the velocity is increased above a critical value, as shown in following figures.



a) Laminar flow b) Turbulent flow c) Transitional flow

We can verify the existence of these laminar, transitional, and turbulent flow regimes by injecting some dye streaks into the flow in a glass pipe, as the

British engineer Osborne Reynolds (1842–1912) did over a century ago. We observe that the dye streak forms a *straight and smooth line* at low velocities when the flow is laminar (we may see some blurring because of molecular diffusion), has *bursts of fluctuations* in the transitional regime, and *zigzags rapidly and disorderly* when the flow becomes fully turbulent. These zigzags and the dispersion of the dye are indicative of the fluctuations in the main flow and the rapid mixing of fluid particles from adjacent layers.

The *intense mixing* of the fluid in turbulent flow as a result of rapid fluctuations enhances momentum transfer between fluid particles, which increases the friction force on the pipe wall and thus the required pumping power. The friction factor reaches a maximum when the flow becomes fully turbulent.

5.1.1 Reynolds Number

The transition from laminar to turbulent flow depends on the *geometry, surface roughness, flow velocity, surface temperature,* and *type of fluid*, among other things. After exhaustive experiments in the 1880s, Osborne Reynolds discovered that the flow regime depends mainly on the ratio of *inertial forces* to *viscous forces* in the fluid. This ratio is called the *Reynolds number* and is expressed for internal flow in a circular pipe as

$$\mathrm{Re} = \frac{\text{Inertial forces}}{\text{Viscous forces}} = \frac{V_{\mathrm{avg}} D}{\nu} = \frac{\rho V_{\mathrm{avg}} D}{\mu}$$

where V_{avg} = average flow velocity (m/s), D = characteristic length of the geometry (diameter in this case, in m), and $\nu = \mu/\rho$ = kinematic viscosity of the fluid (m2/s). Note that the Reynolds number is a *dimensionless* quantity. Also, kinematic viscosity has units m^2/s, and can be viewed as *viscous diffusivity* or *diffusivity for momentum.*

At large Reynolds numbers, the inertial forces, which are proportional to the fluid density and the square of the fluid velocity, are large relative to the viscous forces, and thus the viscous forces cannot prevent the random and rapid fluctuations of the fluid. At *small* or *moderate* Reynolds numbers, however, the viscous forces are large enough to suppress these fluctuations and to keep the

fluid "in line." Thus the flow is *turbulent* in the first case and *laminar* in the second.

The Reynolds number at which the flow becomes turbulent is called the *critical Reynolds number*, Re_{cr}. The value of the critical Reynolds number is different for different geometries and flow conditions. For internal flow in a circular pipe, the generally accepted value of the critical Reynolds number is $Re_{cr} = 2300$.

It certainly is desirable to have precise values of Reynolds numbers for laminar, transitional, and turbulent flows, but this is not the case in practice. It turns out that the transition from laminar to turbulent flow also depends on the degree of disturbance of the flow by *surface roughness, pipe vibrations,* and *fluctuations in the upstream flow*. Under most practical conditions, the flow in a circular pipe is laminar for $Re \lesssim 2300$, turbulent for $Re \gtrsim 4000$, and transitional in between.

6 Flow Patterns and Flow Visualization

While quantitative study of fluid dynamics requires advanced mathematics, much can be learned from *flow visualization*—the visual examination of flow field features. Flow visualization is useful not only in physical experiments, but in *numerical* solutions as well [computational fluid dynamics (CFD)]. In fact, the very first thing an engineer using CFD does after obtaining a numerical solution is simulate some form of flow visualization, so that he or she can see the "whole picture" rather than merely a list of numbers and quantitative data. There are many types of flow patterns that can be visualized, both physically (experimentally) and/or computationally.

6.1 Streamlines and Streamtubes

> A *streamline* is a curve that is everywhere tangent to the instantaneous local velocity vector.

Streamlines are useful as indicators of the instantaneous direction of fluid motion throughout the flow field. For example, regions of recirculating flow and

separation of a fluid off of a solid wall are easily identified by the streamline pattern. Streamlines cannot be directly observed experimentally except in steady flow fields, in which they are coincident with pathlines and streaklines, to be discussed next. Mathematically, however, we can write a simple expression for a streamline based on its definition.

Consider an infinitesimal arc length $d\vec{r} = dx\hat{i}+dy\hat{j}+dz\hat{k}$ along a streamline; $d\vec{r}$ must be parallel to the local velocity vector $\vec{V} = u\hat{i} + v\hat{j} + w\hat{k}$ by definition of the streamline. By simple geometric arguments using similar triangles, we know that the components of $d\vec{r}$ must be proportional to those of $\vec{V}$. Hence the equation for a streamline is,

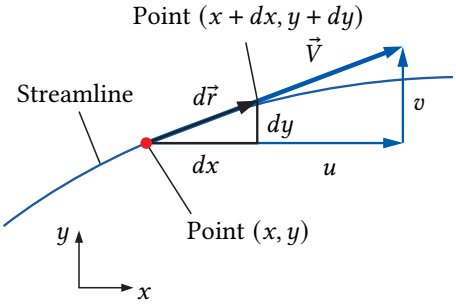
$$\frac{dr}{V} = \frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

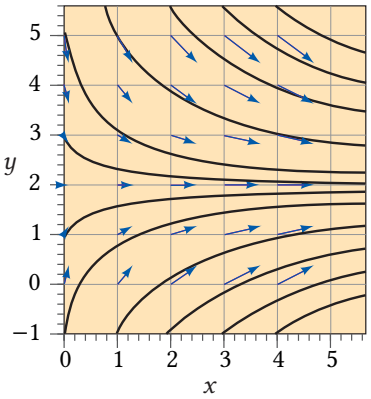
where dr is the magnitude of $d\vec{r}$ and V is the speed, the magnitude of velocity vector $\vec{V}$. Above equation is illustrated in two dimensions for simplicity in the left of following figure. For a known velocity field, we integrate the above equation to obtain equations for the streamlines. In two dimensions, (x, y), (u, v), the following differential equation is obtained for streamline in the xy-plane:

$$\left(\frac{dy}{dx}\right)_{\text{along a streamline}} = \frac{u}{v}$$

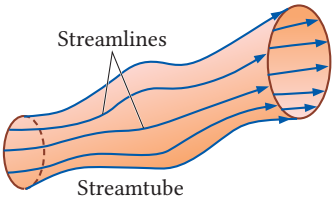
In some simple cases, above equation may be solvable analytically; in the general case, it must be solved numerically. In either case, an arbitrary constant of integration appears. Each chosen value of the constant represents a different streamline. The *family* of curves that satisfy above equation therefore represents streamlines of the flow field.

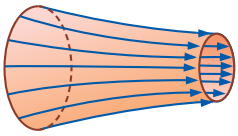
As an example, streamlines (solid black curves) for a steady, incompressible, two-dimensional velocity field given by $\vec{V} = (u, v) = (0.5+0.8x)\hat{i}+(1.5-0.8y)\hat{j}$ are given in right of the following figure. Here the x- and y-coordinates are in meters and the magnitude of velocity is in m/s. Velocity vectors (blue arrows) are superimposed for comparison.



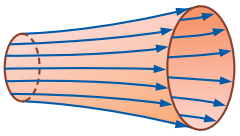


A streamtube consists of a bundle of streamlines (see below figure), much like a communications cable consists of a bundle of fiber-optic cables. Since streamlines are everywhere parallel to the local velocity, fluid cannot cross a streamline by definition. By extension, *fluid within a streamtube must remain there and cannot cross the boundary of the streamtube.* You must keep in mind that both streamlines and streamtubes are instantaneous quantities, defined at a particular instant in time according to the velocity field at that instant. In an *unsteady* flow, the streamline pattern may change significantly with time. Nevertheless, at any instant in time, the mass flow rate passing through any cross-sectional slice of a given streamtube must remain the same. For example, in a converging portion of an incompressible flow field, the diameter of the streamtube must decrease as the velocity increases in order to conserve mass (see below figure *a*). Likewise, the streamtube diameter increases in diverging portions of an incompressible flow (see below figure *b*).





a) Streamtube for a flow which accelerates or converges

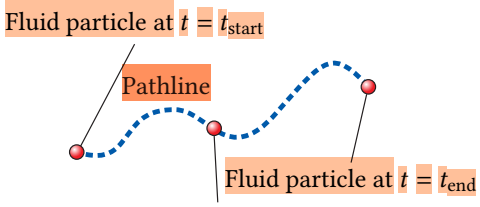


b) Streamtube for a flow which decelerates or diverges

6.2 Pathlines

A *pathline* is the actual path traveled by an individual fluid particle over some time period.

Pathlines are the easiest of the flow patterns to understand. A pathline is a Lagrangian concept in that we simply follow the path of an individual fluid particle as it moves around in the flow field (See the below figure). Thus, a pathline is the same as the fluid particle's material position vector $(x(t), y(t), z(t))$, traced out over some finite time interval. In a physical experiment, you can imagine a tracer fluid particle that is marked somehow—either by color or brightness—such that it is easily distinguishable from surrounding fluid particles. Now imagine a camera with the shutter open for a certain time period, $t_{start} < t < t_{end}$, in which the particle's path is recorded; the resulting curve is called a pathline.



Pathlines can also be calculated numerically for a known velocity field. Specifically, the location of the tracer particle is integrated over time from some starting location $\vec{x}_{start}$ and starting time t_{start} to some later time t. Hence the tracer particle location at time t is,

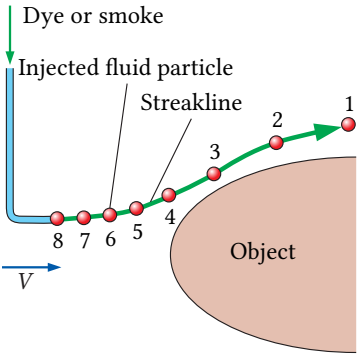
$$\vec{x} = \vec{x}_{start} + \int_{t_{start}}^{t} \vec{V}\, dt$$

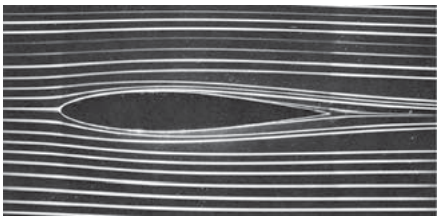
When this equation is calculated for t between t_{start} and t_{end}, a plot of $x(t)$ is the pathline of the fluid particle during that time interval, as illustrated in the above. For some simple flow fields, this equation can be integrated analytically. For more complex flows, we must perform a numerical integration.

If the velocity field is steady, individual fluid particles follow streamlines. Thus, *for steady flow, pathlines are identical to streamlines.*

6.3 Streaklines

A *streakline* is the locus of fluid particles that have passed sequentially through a prescribed point in the flow.





Streaklines are the most common flow pattern generated in a physical experiment. If you insert a small tube into a flow and introduce a continuous stream of tracer fluid (dye in a water flow or smoke in an airflow), the observed pattern is a streakline. Above figure shows a tracer being injected into a free-stream flow containing an object, such as a wing. Labeled tracer particles (1 through 8) were introduced sequentially. The circles represent individual injected tracer fluid particles, released at a uniform time interval. As the particles are forced out of the way by the object, they accelerate around the shoulder of the object, as indicated by the increased distance between individual tracer particles in that region. The streakline is formed by connecting all the circles into a smooth curve. In physical experiments in a wind or water tunnel, the smoke or dye is injected *continuously*, not as individual particles, and the resulting flow pattern is by definition a streakline. In the figure, tracer particle 1 was released at an earlier time than tracer particle 2, and so on. The location of an individual tracer particle is determined by the surrounding velocity field from the moment of its injection into the flow until the present time. If the flow is unsteady, the surrounding velocity field changes, and we cannot expect the resulting streakline to resemble a streamline or pathline at any given instant in time. However, *if the flow is steady, streamlines, pathlines, and streaklines*

are identical. Streaklines produced by colored fluid introduced in a upstream is shown in the above photograph (right side). Since the flow is steady, these streaklines are the same as streamlines and pathlines.

Streaklines are often confused with streamlines or pathlines. While the three flow patterns are identical in steady flow, they can be quite different in unsteady flow. The main difference is that a streamline represents an *instantaneous* flow pattern at a given instant in time, while a streakline and a pathline are flow patterns that have some *age* and thus a *time history* associated with them. A streakline is an instantaneous snapshot of a *time-integrated* flow pattern. A pathline, on the other hand, is the *time-exposed* flow path of an individual particle over some time period.

For a known velocity field, a streakline can be generated numerically. We need to follow the paths of a continuous stream of tracer particles from the time of their injection into the flow until the present time. Mathematically, the location of a tracer particle is integrated over time from the time of its injection t_{inject} to the present time t_{present}. Hence the integrated tracer particle location becomes

$$\vec{x} = \vec{x}_{\text{injection}} + \int_{t_{\text{inject}}}^{t_{\text{present}}} \vec{V}\,dt$$

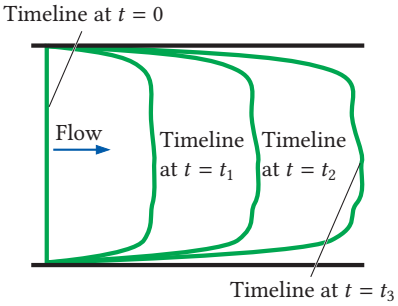
In a complex unsteady flow, the time integration must be performed numerically as the velocity field changes with time. When the locus of tracer particle locations at $t = t_{\text{present}}$ is connected by a smooth curve, the result is the desired streakline.

6.4 Timelines

> A *timeline* is a set of adjacent fluid particles that were marked at the same (earlier) instant in time.

Timelines are particularly useful in situations where the uniformity of a flow (or lack thereof) is to be examined. The following figure illustrates timelines in a channel flow between two parallel walls. Because of friction at the walls, the fluid velocity there is zero (the no-slip condition), and the top and bottom of the timeline are anchored at their starting locations. In regions of the flow

away from the walls, the marked fluid particles move at the local fluid velocity, deforming the timeline. In the example of the following figure, the speed near the center of the channel is fairly uniform, but small deviations tend to amplify with time as the timeline stretches. Timelines can be generated experimentally in a water channel through use of a *hydrogen bubble wire*. When a short burst of electric current is sent through the cathode wire, electrolysis of the water occurs and tiny hydrogen gas bubbles form at the wire. Since the bubbles are so small, their buoyancy is nearly negligible, and the bubbles follow the water flow nicely.

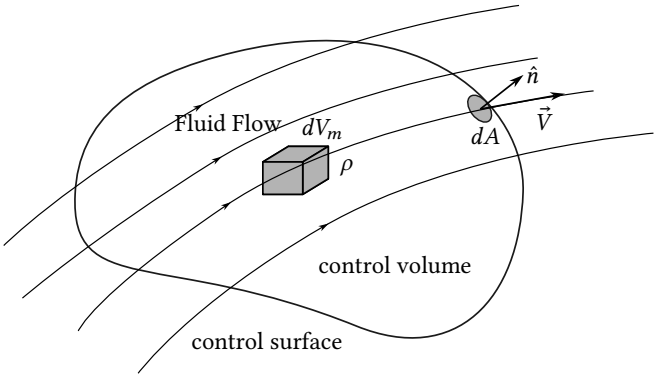


7 Continuity Equation in Integral form

Setting aside nuclear reactions and relativistic effects, mass is neither created nor destroyed. Thus, individual mass elements – molecules, grains, fluid particles, etc., may be tracked within a flow field because they will not disappear and new elements will not spontaneously appear. Through application of the Reynolds transport theorem, we have the following general expression for conservation of mass as applied to a control volume:

$$\int_{\substack{\text{control}\\\text{volume}}} \frac{\partial \rho}{\partial t} dV_m + \oint_{\substack{\text{control}\\\text{surface}}} \rho \vec{V}.\hat{n}\, dA = 0 \tag{1}$$

where ρ is the density of the fluid at each point in the control volume, $\vec{V}$ is the velocity of the fluid at each point on the control surface, $\hat{n}$ is the unit vector



outward normal to the control surface, dV_m is the infinitesimal volume element and dA is the infinitesimal area element. This is the *integral form of the continuity equation* and is valid for both fixed and moving control volumes, provided that the velocity vector is the *absolute* velocity (as seen by a fixed observer). When there are well-defined inlets and outlets, this equation is rewritten as

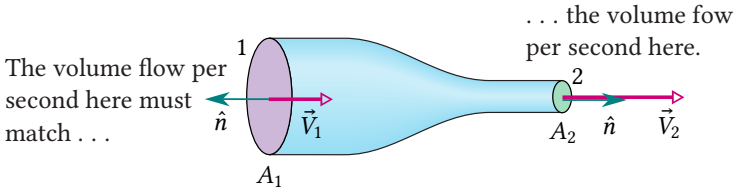
$$\int_{CV} \frac{\partial \rho}{\partial t} dV_m = \sum_{in} \dot{m} - \sum_{out} \dot{m} \tag{2}$$

In words, the net rate of change of mass within the control volume is equal to the rate at which mass flows into the control volume minus the rate at which mass flows out of the control volume. Equation (1) applies to any control volume, regardless of its size.

A simple case in which equation (1) can be easily solved is for the steady flow of an ideal fluid through a tube with varying cross section, like that in the below figure. The flow is toward the right, and the tube segment shown is part of a longer tube. The fluid has average speeds V_1 at the left end (surface 1) of the segment and V_2 at the right end (surface 2). The tube has cross-sectional areas A_1 at the left end and A_2 at the right end.

If the fluid is incompressible (like liquids),

$$\frac{\partial \rho}{\partial t} = 0$$



so that the equation (1) becomes,

$$\oint_{\substack{\text{control}\\\text{surface}}} \rho \vec{V}.\hat{n}\, dA = 0$$

Here there are only two surfaces to be considered for evaluating the above surface integral, the opening through which the fluid enters the control volume and the other opening through which the fluid leaves the control volume. For the present case, the above integral will become,

$$\rho \int_{\text{surface 1}} \vec{V}.\hat{n}\, dA + \rho \int_{\text{surface 2}} \vec{V}.\hat{n}\, dA = 0$$

and if the fluid enters and leaves with normal to the surfaces 1 and 2, then $\vec{V}.\hat{n}$ for surface 1 is $-V_1$ and for surface 2 is V_2. Hence the above equation becomes,

$$-V_1 \int_{\text{surface 1}} dA + V_2 \int_{\text{surface 2}} dA = 0$$

$$-V_1 A_1 + V_2 A_2 = 0$$

$$A_1 V_1 = A_2 V_2 \tag{3}$$

This relation between speed and cross-sectional area is the equation of continuity for the flow of an ideal fluid through a tube making its velocity normal to the entry and exit surfaces. It tells us that the flow speed increases when we decrease the cross-sectional area through which the fluid flows. You may have noticed that you can increase the speed of the water emerging from a garden hose by partially closing the hose opening with your thumb.

We can rewrite equation (3) as

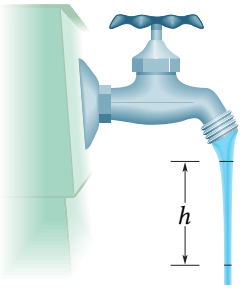
$$R_V = AV = \text{a constant}$$

in which R_V is the volume flow rate of the fluid (volume past a given point per unit time). Its SI unit is the cubic meter per second (m^3/s). If the density ρ of the fluid is uniform, we can multiply the above equation by that density to get the mass flow rate R_m (mass per unit time):

$$R_m = \rho R_V = \rho AV = \text{a constant}$$

The SI unit of mass flow rate is the kilogram per second (kg/s). This equation says that the mass that flows into the tube segment at each second must be equal to the mass that flows out of that segment at each second.

Example 5. The below figure shows how the stream of water emerging from a faucet "necks down" as it falls. This change in the horizontal cross-sectional area is characteristic of any laminar (non-turbulent) falling stream because the gravitational force increases the speed of the stream. Here the indicated cross-sectional areas are $A_i = 1.2\,cm^2$ and $A_f = 0.35\,cm^2$. The two levels are separated by a vertical distance $h = 45\,mm$. What is the volume flow rate from the tap?



The volume fow per second here must match ... A_i, V_i

A_f, V_f ... the volume fow per second here.

Solution. The volume flow rate through the higher cross section must be the same as that through the lower cross section. From continuity equation we have

$$A_i V_i = A_f V_f$$

where V_i and V_f are the water speeds at the levels corresponding to A_i and A_f. From the equations of motion, since the water is falling freely with acceleration g,

$$V_f^2 = V_i^2 + 2gh$$

Eliminating V_f between above two equations and solving for V_i, we obtain

$$\begin{aligned} V_i &= \sqrt{\frac{2ghA_f^2}{A_i^2 - A_f^2}} \\ &= \sqrt{\frac{(2)(9.8\,\text{m/s}^2)(0.045\,\text{m})(0.35\,\text{cm}^2)^2}{(1.2\,\text{cm}^2)^2 - (0.35\,\text{cm}^2)^2}} \\ &= 0.286\,\text{m/s} = 28.6\,\text{cm/s}. \end{aligned}$$

Hence the the volume flow rate R_V is then

$$\begin{aligned} R_V &= A_i V_i = (1.2\,\text{cm}^2)(28.6\,\text{cm/s}) \\ &= 34\,\text{cm}^3/\text{s}. \end{aligned}$$

□

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[4] Jearl Walker, *Halliday & Resnick Fundamentals of Physics,* 11th ed., John Wiley & Sons, Inc, 2018.